ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 21-Sep-2026

Site: Watermark Test Site | Location: Shivamogga, Karnataka | Engine: 100% Old Code Algorithm Match

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
9.99 kW	**16,569 kWh**	**Rs. 480,450**	**4Y 7M**
10.45 kWp Installed	CUF 18.1% | PR 75%	Rs. 4.80 Lakhs	Profit: Rs. 48.79L

[SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

Company / Customer Name	Watermark Test Site
Location / Address	Shivamogga, Karnataka
Contact Email Address	test@voltroof.com
Design Path Selected	Rooftop Area Analysis
System Target Capacity	9.99 kW
Total Installed Array Rating	10.45 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	19 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	1200.00 Sq.Feet (111.48 sq.m)
Utilized Usable Rooftop Area	999 Sq.Feet (92.81 sq.m)
First Year Annual Generation	16,568.88 kWh
25-Year Cumulative Generation	368,125.12 kWh
Estimated Payback Period	4 Years & 7 Months
Post-Payback Profitability (25 Yrs)	Rs. 4,879,278.67 (INR 48.79 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 2,319,468.04 (INR 23.19 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 301 Tonnes CO2 | Tree Equivalence: 481 Mature Trees | Green Energy: 100% Clean LT Power

Based on Central Electricity Authority (CEA) grid emission factor (0.82 kg CO2/kWh) and standard forestry carbon sequestration.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Site 1 (Rooftop Array)
Gross Rooftop Area	1200.00 Sq.Feet (111.48 sq.m)
Utilized Effective Rooftop Area	999 Sq.Feet (92.81 sq.m)
Building Plant Target Capacity	9.99 kW
Solar PV Module Rating	550 W Mono PERC / TOPCon Half-Cut
Number of Solar Modules	19 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Inverter Rating	11 kW, 415V, 3-Ph, MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications

Core inverter rating, MPPT topology, and operating parameters

Selected Inverter Rating	11 kW Grid-Tied Solar Inverter
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx16 Sq.mm
DC Array Isolator Switch	50A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	15.30 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx16 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550 W mono PERC / TOPCon	19 Nos.	Rs. 190,000
02	On-grid inverter	11 kW, three-phase	1 No.	Rs. 121,000
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 45,000
04	DC solar cable	6 sq.mm copper, PV-rated	110 m	Rs. 9,350
05	AC cable	16 sq.mm copper	60 m	Rs. 14,400
06	MC4 connectors	PV-rated connector pairs	19 pairs	Rs. 5,700
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 11,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 11,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 5,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 15,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 45,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 480,450.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 5,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 5,000 (Refundable utility escrow)

STATUTORY COMPLIANCE & NET-METERING POLICY

All electrical installations conform to CEA (Technical Standards for Connectivity of Distributed Generation Resources) Regulations 2013 and State Grid Code. Net-metering bi-directional billing operates under applicable KERC / State SERC LT Solar Tariff orders.

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 480,450	**4Y 7M**	**Rs. 48.79L**	**Rs. 23.19L**
12-Item BOM Basis	Break-Even Year	Rs. 4,879,279	Alternative Asset

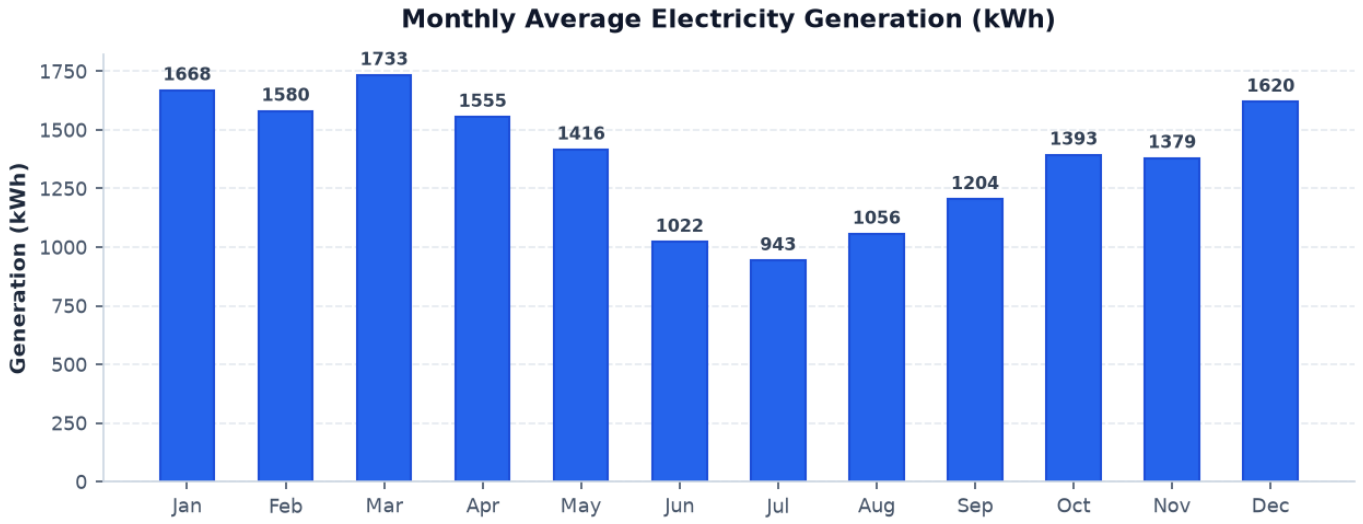
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	16,569	Rs. 8.05	133,379	5,994	127,385	127,385
02	16,403	Rs. 8.45	138,607	5,994	132,613	259,998
03	16,239	Rs. 8.87	144,041	5,994	138,047	398,046
04 *	**16,077**	**Rs. 9.32**	**149,835**	**5,994**	**143,841**	**541,887**
05	15,916	Rs. 9.78	155,658	5,994	149,664	691,552
06	15,757	Rs. 10.27	161,823	5,994	155,829	847,380
07	15,599	Rs. 10.78	168,160	5,994	162,166	1,009,547
08	15,443	Rs. 11.32	174,818	5,994	168,824	1,178,371
09	15,289	Rs. 11.89	181,784	5,994	175,790	1,354,161
10	15,136	Rs. 12.48	188,897	5,994	182,903	1,537,064
11	14,985	Rs. 13.11	196,448	5,994	190,454	1,727,518
12	14,835	Rs. 13.76	204,126	5,994	198,132	1,925,650
13	14,686	Rs. 14.45	212,219	5,994	206,225	2,131,875
14	14,540	Rs. 15.18	220,710	5,994	214,716	2,346,591
15	14,394	Rs. 15.93	229,299	5,994	223,305	2,569,895
16	14,250	Rs. 16.74	238,548	5,994	232,554	2,802,450
17	14,108	Rs. 17.57	247,872	5,994	241,878	3,044,328
18	13,967	Rs. 18.45	257,684	5,994	251,690	3,296,018
19	13,827	Rs. 19.37	267,828	5,994	261,834	3,557,853
20	13,689	Rs. 20.34	278,428	5,994	272,434	3,830,287
21	13,552	Rs. 21.36	289,466	5,994	283,472	4,113,759
22	13,416	Rs. 22.43	300,927	5,994	294,933	4,408,692
23	13,282	Rs. 23.55	312,794	5,994	306,800	4,715,492
24	13,149	Rs. 24.73	325,182	5,994	319,188	5,034,680
25	13,018	Rs. 25.96	337,942	5,994	331,948	5,366,629

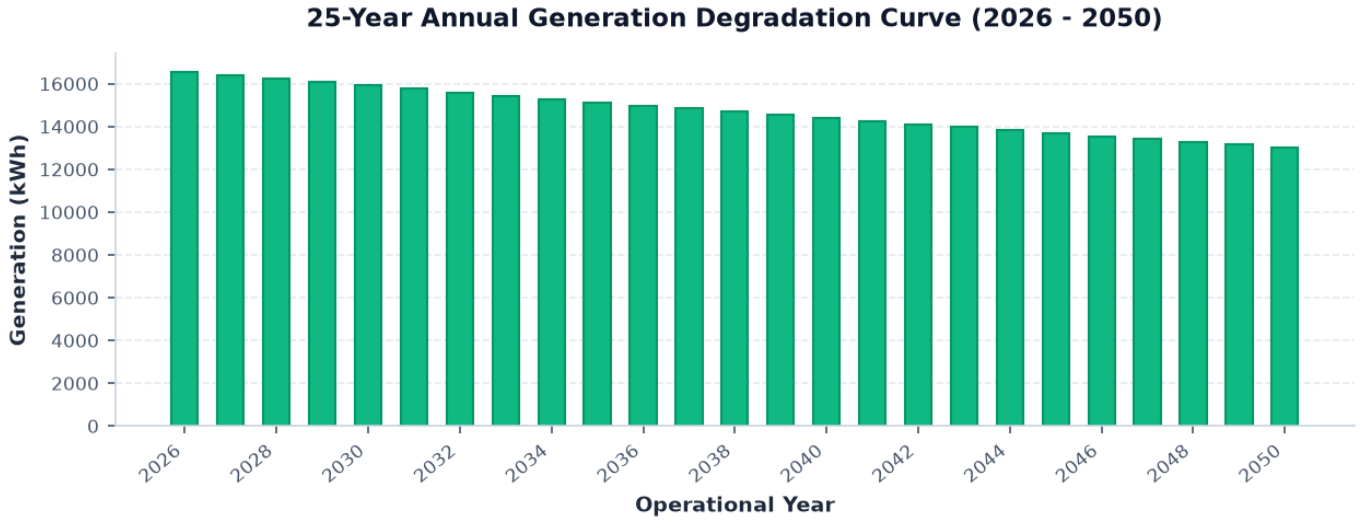
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insulation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1659 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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ON-GRID SOLAR PV SYSTEM

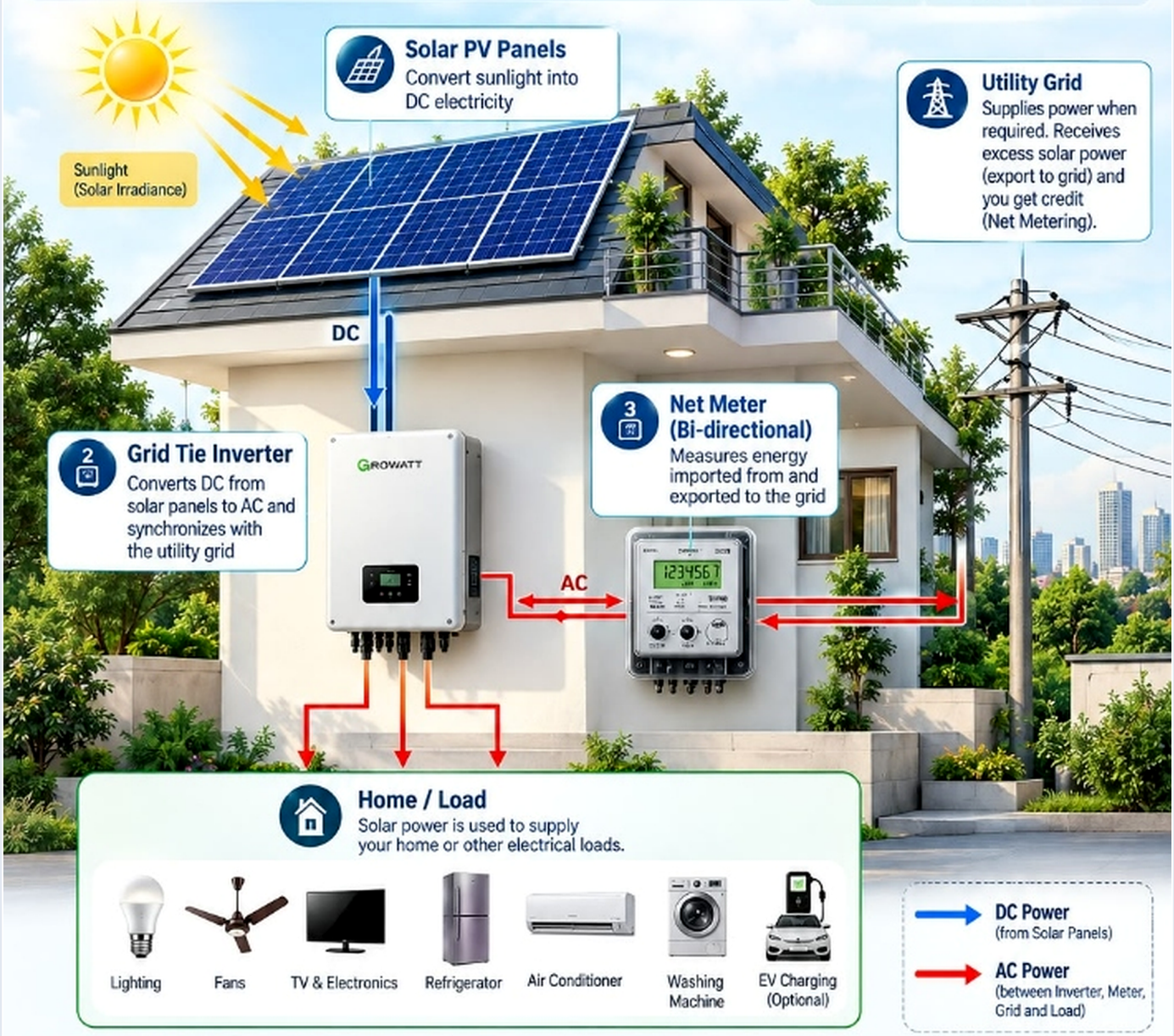
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

- **DC Power** (from Solar Panels)
- **AC Power** (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow